Pablo Cuellar

Professor Obaid Junaid

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What is firewall and how does it help to protect the network?

Abstract

As global infrastructure increasingly migrates toward digital frameworks, the frequency of exploits and vulnerabilities has accelerated, often outpacing traditional defensive countermeasures. To combat these threats, cybersecurity professionals rely on the firewall as a pivotal security appliance designed to shield networks from unauthorized intrusions. This paper examines the technical mechanisms through which firewalls protect network integrity, specifically analyzing packet filtering, stateful inspection, and the implementation of standardized security rules. While firewalls serve as an essential first line of defense, this research concludes that they are most effective when integrated into the broader, multi-layered security strategy rather than functioning as a singular solution.

Introduction

In the early days of networked computing, security was often an afterthought, predicated on a “trusted” internal environment. In response, the firewall emerged as the fundamental “gatekeeper” of the network, designed to enforce a security policy at the boundary between two networks of different trust levels. Because everything is connected, hackers have more ways to get into our private information than ever before. Cybersecurity is a constant race. New

vulnerabilities pop up every day, and a lot of times, the "bad guys" find them before the "good guys" do. A firewall as a security device that sits between your private network and the rest of the internet, it acts like a filter (Cisco). It looks at all the data trying to enter your network and decides if it's safe or if it looks like a threat. If you don't have one, your network is basically an open door. But having one isn't enough you have to know how to set it up. In this paper, I'm going to walk through the history of how these devices started, how they use rules to make decisions, and how they're using things like AI to get even smarter. Firewalls aren't perfect, but they're the first thing you need if you want to keep your data safe.

As organizational networks have grown in complexity incorporating remote work, IoT devices, and multi cloud environments the boundary has become increasingly porous. This evolution has forced firewalls to move beyond simple port and IP based filtering. Modern firewalls must now operate at the application layer of the OSI model, performing complex tasks such as SSL/TLS decryption and protocol normalization to identify malicious payloads hidden within legitimate traffic (Dhiman et al., 2024). This paper will provide a comprehensive literature review of those technologies, assessing their effectiveness against the modern adversarial tactics and their continued relevance in an era where the traditional network boundary is effectively disappearing.

Literary Review

To understand how firewalls work today, you have to look at where they started. According to Palo Alto Networks, firewalls have gone through a few different generations as technology got more complex. Back in the late 1980s, the first firewalls were pretty basic. They were called "packet filters." Think of these, like security guards with a very simple checklist,

they only looked at the envelope of the data, where it was coming from and where it was going. They didn't really care what was inside the message.

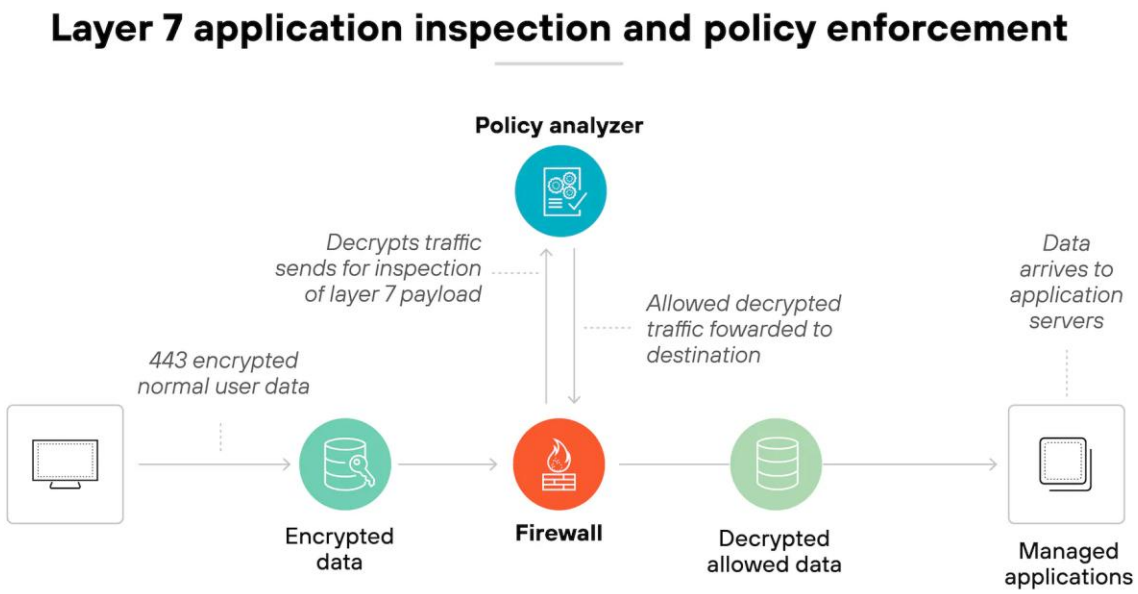
In the 1990s, things got better with stateful inspection, this was a big jump because the firewall started remembering conversations. If you started a connection with a website, the firewall would remember that you were the one who started the connection, when the website sent data back, the firewall knew it was part of an ongoing connection and let it through. This made things so much safer because hackers couldn't just send random data into your network without you asking for it first ("The History of Firewalls | Who Invented the Firewall?").

The most basic job of a firewall is to look at the packets, when you send anything over the internet, like an email or a photo, it gets broken down into tiny little pieces called packets. Each packet has a header that says things like the IP address it's coming from and the port it's trying to use. Palo Alto Networks explains that firewalls use rules to decide what to do with these packets, these rules are basically a big "To do" list for the firewall. A typical rule looks like this: "If a packet is coming from [IP Address A] and trying to go to [Port 80], then [Allow] it." The firewall reads this list from the top to the bottom; once it finds a rule that matches the packet, it stops looking and does what the rule says. One of the most important concepts here is the "Implicit Deny" this is a rule at the very bottom of the list that basically says, "If a packet didn't match any of the safe rules above, block it." It's a "better safe than sorry" approach. If the firewall doesn't know what it is, don't let it in.

Sometimes you actually want to let people into your network from the outside. For example, if you are running a web server or a game server, people need a way to connect to you. This is where port forwarding comes in. You are basically telling the firewall, "Hey, is someone knocks on port 80, send them to my web server." While this is useful, it is also risky. Every port

you open is like another window in your home being left open, someone might try to crawl through it. That is why managing these ports is a huge part of a cybersecurity expert's job, you only want to open exactly what you need and nothing more.

As hackers got smarter, basic firewalls were not enough anymore, they started hiding bad code inside "safe" looking packets. This lead to what Palo Alto calls Next Generation Firewalls, these are much more powerful because they use Deep Packet Inspection. Instead of just looking at the header of the data an NGFW actually opens the packets and reads the data inside. It can tell the difference between someone using Facebook for work and someone trying to download a virus disguised as a Facebook game. It also has an Intrusion Prevention System (IPS) built right in, which means it can spot an attacker while it is happening and shut it down immediately.



For a long time, everyone focused on the "perimeter." The idea was that if you had a really strong firewall at the front door, the inside of your network was safe but as Illumio points out, that is not really true anymore. If a hacker gets past that front door, maybe through a

phishing email or a stolen password, they are inside the network. In a traditional setup once they are in, they can move around by escalating their privileges and they can gain access to everything.

This is where micro segmentation come into play, instead of just one big firewall at the edge of your network, you basically put a tiny firewall around every single device or "segment" of the network. Illumio uses a great analogy for this: It's like a submarine. A submarine is built with different compartments. If one part of the sub gets a leak, you shut the doors to that section. The rest of the ship stays dry and keeps floating. In a network, micro segmentation does the same thing (Illumio). If a hacker gets into one employee's laptop, the tiny firewall around the database and the server realizes the laptop should not be talking to them, they shut the access down. This stops what we call lateral movement, where a small hack turns into a massive data breach. It was a massive shift in thinking instead of trusting everyone inside, you trust nobody.

The newest thing in the world of firewalls is Artificial intelligence. Traditionally, firewalls worked off signatures. It is like having a book of mugshots, if a packet looks like a known virus, the firewall blocks it. The problem is that hackers are constantly making new viruses that are not in the book yet, those are called zero-day attacks. According to Huawei and Metomic, AI firewalls do not just look at the signatures, they look at the behavior. It is like a bouncer at a club who is not just looking for people on a banned list but is also watching for someone acting suspicious (Shui; "A Guide to AI Firewalls and How They Can Protect Your Data | Metomic"). If a user who usually only uploads 1 GB of data a day suddenly tries to upload 50 GB to a server in the middle of the night, an AI firewall sees that "weird" behavior and stops it immediately. It does not wait for a signature to tell it that something is wrong.

Metomic also talks about a very specific modern problem: employees using AI tools like ChatGPT. It's easy for someone to accidentally paste sensitive company data into a prompt to get help with a project. An AI aware firewall can actually spot sensitive data (like a credit card number or a secret code) and block it from ever leaving the network. It is basically protecting the company from its own mistakes.

Huawei takes this step further with their "HiSecEngine." The idea here is speed. Hackers are using automated tools to attack networks, we need automated tools to defend them, when an AI firewall spots a threat, it can update its own rules in seconds across the entire network, a human could never type that fast. By the time a security expert gets an alert on their phone, the AI firewall has already identified the threat, blocked the IP address, and closed the "hole" in the system.

If you want to see what happens when a firewall rules aren't up to date, you have to look at the 2014 Sony Pictures hack. This is the "big one" that everyone in cybersecurity still talks about. A group calling itself the "Guardian of Peace" basically walked into Sony's network, stole nearly 100 terabytes of data including unreleased movies and private emails and their wiped their servers clean (FBI).

So, how does a giant company like Sony let that happen? They definitely had firewalls. They had a security team, but the Sony hack is the perfect example of why a "perimeter only" defense does not work anymore. The biggest issue at Sony, at that time, was their internal network was "flat" (Sanchez). Imagine a giant warehouse with a massive, high-tech lock on the front door, but once you get inside, there are no other locked doors or walls. Everything, the payroll data, the movie scripts, the employee emails, was all just sitting in one big room.

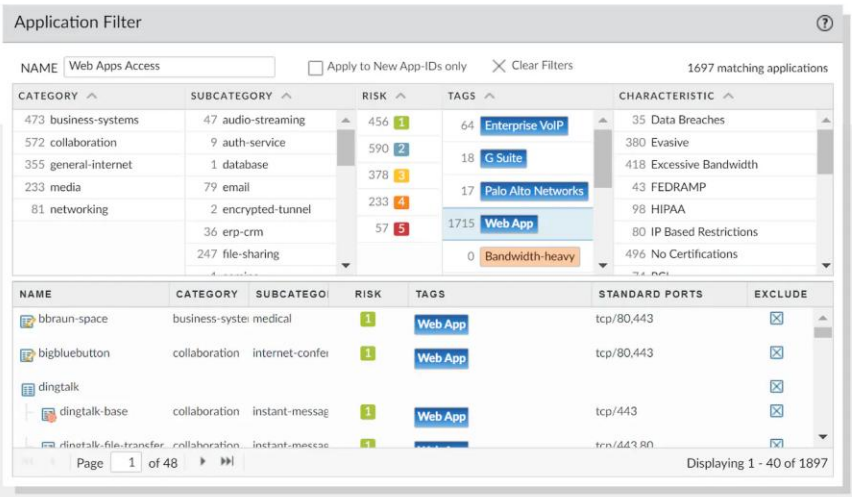
Once the hackers got past the perimeter firewall (likely through a simple phishing email that stole someone's password), they were inside Sony's network. Because Sony was not using the kind of micro segmentation we talked about earlier from Illumio, the hackers could move laterally across the network with zero resistance. The firewalls at the edge of the network did not see anything wrong because as far as they were concerned that traffic was already inside the "trusted zone" Sony's Network.

Another major failure was on the outbound side of the firewall, most people think of firewalls as things that keep hackers out, but a huge part of their job is also watching what is going out of the network. The hackers spent weeks, even months, slowly downloading 100 Terabytes of data to their own servers, that is a massive amount of traffic. If Sony had been using a Next Generation Firewall (NGFW) with better visibility to the packets going in and out of the network. A smart firewall would have asked itself, "Wait, why is a computer is the accounting department sending terabytes of data to a random server in another country?" and shut down the connection, but because the firewall rules were likely too loose on outgoing traffic, the hackers just walked right out the door with all the loot.

If we look at the Sony Hack through the lens of the tech, we have today, the story would be different:

- Micro segmentation: If Sony had compartmentalized their network like the submarine analogy, the hackers might have gotten into one department, but they would not have been able to reach the movie servers or the private emails.
- AI Behavioral Analysis: An AI Firewall would have flagged the massive data download as a "behavioral anomaly" and notified the security experts and block it instantly, even without knowing it was a hack.

- App-ID: An NGFW could have seen that the hackers were using specific tools or protocols to move the data and blocked those specific applications while letting regular work continue ("What Is a Next-Generation Firewall (NGFW)? A Complete Guide").



The firewall identifies and classifies traffic by application—not just by port—using attributes like category, risk level, and behavioral characteristics.

The Sony hack proved that the "mighty" firewall needs to be everywhere, not just at the front door, it needs to be watching every internal move and every piece of data trying to leave the network.

To really get why firewalls are so important, you must understand what they are looking at. Earlier, I mentioned packets and headers, let's break that down. When you send a piece of data, it is not just the message itself traveling through the network and the Word Wide Web. It is wrapped in layers of information, sort of like a nesting doll. The most common way firewalls look at this is through something called the 5-tuple. It sounds like a fancy math term, but it is actually just five specific pieces of information that the firewalls use to identify and manage network traffic flows:

1. Source IP Address: This is your computer's "home address" on the internet. It tells the web server where to send the data back to.
2. Destination IP Address: This is the address of the server you're trying to talk to. Without this, the packet wouldn't know where to go.
3. Source Port: This is a "random" door your computer opens to let the data out. Your computer usually picks a high number that isn't being used by anything else.
4. Destination Port: This is the most important part of the firewall. It tells the server what kind of "service" you want. If you're browsing a secure website, you're almost always using Port 443 (HTTPS). If the firewall sees a request for Port 443, it usually knows it's just someone browsing the web. But if it sees a request for a "scary" port like Port 23 (Telnet), it might get suspicious.
5. Protocol This is the "language" or the set of rules the data is using to travel. Most web traffic uses TCP (Transmission Control Protocol) because it's reliable—it makes sure every piece of the data arrives in the right order. If the packet is for something like a live video stream where speed is more important than being perfect, it might use UDP (User Datagram Protocol) instead ("Understanding the 5-Tuple in Network Communication").

Understanding the 5-tuple is great, but the real "magic" happens when the firewall uses those five pieces of info to make a split-second decision. Think of the firewall like a security guard at a high-security building who has a very specific "Guest List." Every single packet that hits the firewall is like a person trying to walk through the door. The firewall stops them, looks at their "ID", the 5-tuple, and checks the Guest List, the Firewall Rules.

Step 1: The Arrival and Identification: When a packet arrives, the firewall immediately "freezes" the packet at the interface. It pulls out those five pieces of info: where it's from, where

it’s going, the ports being used, and the protocol. At this point, the packet hasn't actually entered the network yet; it’s sitting in a "waiting room."

Step 2: Scanning the Rule List: The firewall starts at the very top of its rule list. This is important: firewalls read from Top to Bottom. It looks at Rule #1. If the packet’s 5-tuple doesn’t match Rule #1, it moves to Rule #2. It keeps going until it finds a match.

Step 3: The Action (Allow vs. Drop): Once it finds a match, it stops looking and takes the action listed in that rule. There are usually two main choices:

- **Allow (or Permit):** The packet is verified and let through the "door."
- **Drop (or Deny):** The firewall silently ignores the packet. It’s like the packet never existed.
- **Reject:** This is similar to a drop, but the firewall sends a message back to the sender saying, "Sorry, you're not allowed in." (Most security pros prefer "Drop" because it doesn't give hackers any information back).

Example 1: The "Allow" Scenario Let's say you're trying to check your school email. Your computer (192.168.1.10) sends a packet to the email server (10.0.0.50) on Port 443 using TCP.

- The firewall looks at its list. Rule #5 says: **"Source: 192.168.1.0/24 | Destination: Any | Port: 443 | Action: ALLOW."**
- Because your computer is in that 192.168.1.0 range and you’re using Port 443, it’s a match. The firewall opens the gate and lets you check your mail.

Example 2: The "Drop" Scenario Now, imagine a hacker at IP address 5.5.5.5 is trying to "brute force" their way into your network's file server (10.0.0.20) using Port 22 (the port for SSH, which is used for remote control).

- The firewall starts at the top of the list. It doesn't find any rule that says, "Allow 5.5.5.5."
- It gets to the very bottom of the list to the **"Implicit Deny"** rule. This rule basically says: **"Source: Any | Destination: Any | Port: Any | Action: DROP."**
- Since the hacker's packet didn't match any of the "Safe" rules above it, it hits this final wall. The firewall drops the packet, and the hacker is stuck outside.

By using this simple "Top-Down" logic, the firewall can process millions of packets every second without getting confused ("What Are Firewall Rules? | Firewall Rules Explained"). It's all about the rules. If the rules are written well, the "good" packets get through instantly, and the "bad" ones never even get a chance to knock on the door.

One of the biggest mistakes a network admin can make is using an "Any-Any" rule. In the firewall world, "Any" is a wildcard. It means everything, an "Any-Any" rule basically means, "Let any traffic from any location go to any destination." Why would someone do this? Usually it is for troubleshooting, if a network is not working and you are frustrated as a network administrator, you might temporarily turn on an "Any-Any" rule just to see if the firewall is the problem, but if you forget to turn it off, you have basically taken the lock off the front door and invited the whole world in. This is exactly how many hacks happen not because the firewall was not strong enough, but because someone left a rule wide open by mistake.

So far, we've mostly talked about firewalls that protect the whole network. But there's a specific kind called a Web Application Firewall, or WAF. If a regular firewall is the security

guard at the front gate of a building, a WAF is the specialized guard standing right in front of the safe. In technical terms, this means the WAF operates at **Layer 7** of the OSI model—the Application Layer. This is the level where the actual "web" stuff happens, like the text you type into a search bar or the photos you upload to a profile. A regular firewall might let a packet through because it is coming from a "safe" IP address on a "safe" port (like Port 443). But a WAF looks at what is inside that packet. It's looking for sneaky attacks that are designed to trick a website's code.

One of the coolest things about a WAF is that it usually acts as a Reverse Proxy. In a normal setup, a user connects directly to a server. But with a WAF, the user connects to the WAF instead, it checks the request, and if it is safe, it passes it along to the server. According to Cloudflare, this is a huge advantage because the "real" server stays hidden behind the WAF. The internet only sees the WAF's face, not the servers. This makes it much harder for a hacker to attack the server directly because they don't even know exactly where it is or what it looks like (Cloudflare).

Most WAFs are built specifically to stop a list of threats called the **OWASP Top 10**. This is a list that cybersecurity experts keep of the ten most dangerous ways to attack a website. The two biggest ones that a WAF handles are SQL Injection (SQLi) and Cross-Site Scripting (XSS).

- **SQL Injection:** This is when a hacker tries to type database commands into a website's input box (like a login or a search bar). If the website isn't protected, those commands can tell the database to "dump" all its secrets, like every user's password. A WAF sees these commands and blocks them before they ever reach the database.

- **Cross-Site Scripting (XSS):** This is when an attacker tries to inject a malicious script into a webpage that other people are going to see. If you've ever seen a "weird" link on a social media site that automatically posts itself on your profile when you click it, that's often XSS. The WAF cleans these scripts out of the traffic, so they never reach the other users.

Another reason WAFs are so popular is something called Virtual Patching. Imagine a new vulnerability that is discovered in the software your website uses (like WordPress). Normally, your developers would have to drop everything, find the bug, and write a "patch" for it. That takes time. Cloudflare explains that a WAF can protect you during that gap. The WAF company can write a new rule in minutes that blocks any traffic trying to use that specific vulnerability. Even if your website still has the "hole" in its code, the WAF is standing in front of it, blocking anyone who tries to crawl through. It's like putting a piece of plywood over a broken window until the glass repair person can get there ("What is a WAF?").

I've called the firewall "mighty," and it is, but we have to be honest, it can't stop everything. The weakest point in any security system isn't the hardware it's the people using it. If you bet everything on just your firewall, you're setting yourself up for a bad time. In the industry, we use a strategy called "Defense in Depth." Think of it like a castle. You don't just have a moat you have high walls, a drawbridge, guards on the towers, and a safe inside the basement. The idea is that if a hacker manages to jump over the moat (the firewall), they still have to deal with the walls and the guards. According to Fortinet, this layering is necessary because hackers are constantly finding ways to sidestep the firewall entirely (Fortinet, "What Is Defense in Depth? Defined and Explained").

One of the easiest ways for a hacker to get past a firewall is to simply walk through the front door using a stolen password. A firewall sees a correct username and password and thinks, "Okay, this person is allowed in." It doesn't know that the person typing isn't actually the employee. This is where Multi-Factor Authentication (MFA) comes in. MFA adds a second lock that the hacker doesn't have the key to like a code sent to your phone or a fingerprint scan. Even if the firewall lets them knock on the door, they can't get in without that second piece of ID. It's one of the most effective ways to stop credential-theft attacks. But if a hacker does get inside maybe they plug a bad USB drive into a laptop and the firewall at the edge of the network can't see them. This is where we need Endpoint Detection and Response (EDR). As Fortinet explains, EDR is like having a security camera and a guard on every single computer (the "endpoints"). EDR tools watch for weird behavior on the actual laptop or server. If a computer suddenly starts encrypting all its files (which is what ransomware does), the EDR realizes something is wrong and quarantines that computer, cutting it off from the rest of the network before the virus can spread (Fortinet, "What Is EDR - Endpoint Detection and Response?")

The biggest hole in any security system isn't the code; it's the people. You can have the most expensive firewall in the world, but it won't stop an employee from being tricked over a phone call or a text message this is called Social Engineering. A hacker might call an employee pretending to be from the IT department and say, "Hey, we're doing a security update, I need you to disable your firewall for a second so I can run a test." If the employee does it, the hacker is in. This is why Security Awareness Training is so important. We call this building a "Human Firewall." TechBrain explains that teaching people how to spot a fake email or a suspicious phone call is just as important as the hardware you buy (Stewart). If your people know the tricks, the hackers have a much harder time getting a foot in the door.

Conclusion

As I've shown throughout this paper, the firewall has evolved from a simple "security guard with a clipboard" into a high-tech, AI-driven defense system. From the early days of basic packet filtering to the modern world of micro-segmentation and behavioral analysis, firewalls have stayed relevant because they adapted to the threats.

The Sony hack taught us that a "flat" network with a single perimeter firewall isn't enough anymore. Today, we need "Defense-in-Depth" a strategy where firewalls are everywhere, watching every internal move and every piece of data trying to leave the building. Whether you’re a student just starting out in cybersecurity or a professional managing a massive data center, the lesson is the same: the firewall is your best friend, but you have to know how to use it. You have to manage your rules, keep your ports closed, and never stop watching the "doors." The digital world is only getting more dangerous, but with a properly configured firewall and a "trust nobody" mindset, we can keep our networks safe from the next big hack.

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